Geometric Instruments: Worked Solutions

MAIS Physics & Astronomy Club (PAC) · Reasoning, derivations and checks

10 worked solutions · independent explanations · consistency checks

Make a genuine attempt before reading. Each solution explains how to choose the method, carries units through the calculation, and checks the result. The exercises are original MAIS PAC teaching problems, not official IOAA questions or marking schemes. Parameters are supplied locally; an exam's constants take precedence.

Solution index

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Solution 1 — Scale drawing. Return →

The physical length is $3.5 \times 10 = 35$ m. **Check:** the scale is a simple ratio; no trigonometry is needed.

Solution 2 — Small angle. Return →

Convert 1° to radians: 0.01745. The size is $100 \times 0.01745 = 1.75$ m. **Check:** a 1° angle is small, so the approximation is excellent.

Solution 3 — Shadow height. Return →

The triangles are similar, so $h/12 = 1/0.8$. Thus $h = 15$ m. **Check:** the tree's shadow is 15 times the stick's shadow, so its height is 15 times the stick's height.

Solution 4 — Triangulation. Return →

The third angle is $180 - 60 - 70 = 50°$. The side opposite 70° is the distance from the 60° endpoint to the target. By the sine rule, $d = 100 \sin 70° / \sin 50° = 122.7$ m. **Check:** the larger angle faces the longer side.

Solution 5 — Position angle components. Return →

$\Delta\alpha \cos\delta = 10 \sin 45^\circ = 7.07''$ and $\Delta\delta = 10 \cos 45^\circ = 7.07''$. **Check:** at 45° the east and north components are equal.

Solution 6 — Plate scale. Return →

The separation is $15 \times 2 = 30''$. The uncertainty is $0.5 \times 2 = 1''$. **Check:** the same ruler error produces a smaller angular error on a finer-scale chart.

Solution 7 — Parallax. Return →

$d = 1/0.1 = 10$ pc. In light-years this is $10 \times 3.26 = 32.6$ ly. **Check:** smaller parallax means larger distance.

Solution 8 — Angular diameter. Return →

Convert 0.5° to radians: 0.008727. The diameter is $384000 \times 0.008727 = 3350$ km. **Check:** the accepted lunar diameter is about 3474 km; the difference comes from the rounded angular diameter and distance.

Solution 9 — RA conversion. Return →

One second of RA is $15''$ of angle along the equator. At declination 60°, the east–west separation is $15'' \cos 60^\circ = 7.5''$. **Check:** near the pole, RA lines converge, so the same RA difference corresponds to a smaller sky angle.

Solution 10 — Design a measurement. Return →

Measure the horizontal distance d from the building and the elevation angle θ to the top. The height is $h = d \tan\theta$ plus the instrument height above the ground. Reduce uncertainty by measuring from farther away so the angle is not too close to 90°, repeating the angle measurement, and ensuring the tape is horizontal. State the protractor resolution and propagate it through $\tan\theta$.